Fundamental Frequency / Fundamental Period of Continuous Time Signals

To identify the period T, the frequency $f = \frac{1}{T}$ or the angular frequency

$$w = 2\pi f = 2\pi/T$$

of a given sinusoidal or complex exponential signal, it is always helpful to write it in any of the following forms

$$\sin(wt) = \sin(2\pi f t) = \sin(2\pi t/T)$$

The fundamental frequency of a signal is the Greatest Common Divisor (GCD) of all the frequency components contained in a signal and equivalently, the fundamental period is the Least Common Multiple (LCM) of all individual periods of the components.

Example 1.a: Fundamental Frequency of Continuous Time Signals

Find the fundamental frequency of the following continuous time signal

$$x(t) = cos\left(\frac{10\pi}{3}t\right) + sin\left(\frac{5\pi}{4}t\right)$$

The frequencies and periods of the two terms are, respectively,

$$w_1 = \frac{10\pi}{3}, f_1 = \frac{5}{3}, T_1 = \frac{3}{5} \text{ and } w_2 = \frac{5\pi}{4}, f_2 = \frac{5}{8}, T_2 = \frac{8}{5}$$

The fundamental frequency f_0 is the GCD of $f_1 = 5/3$ and $f_2 = 5/8$

$$f_0 = GCD\left(\frac{5}{3}, \frac{5}{8}\right) = GCD\left(\frac{40}{24}, \frac{15}{24}\right) = \frac{5}{24}$$

Example 1.b: Fundamental Period of Continuous Time Signals

Alternatively, the period of the fundamental T_0 is the LCM of $T_1 = \frac{3}{5}$ and $T_1 = \frac{8}{5}$

$$T_0 = LCM\left(\frac{3}{5}, \frac{8}{5}\right) = \frac{24}{5}$$

Now we get $w_0 = 2\pi f_0 = \frac{2\pi}{T_0} = \frac{5\pi}{12}$ and the signal can be written as

$$x(t) = cos\left(8\frac{5\pi}{12}t\right) + sin\left(3\frac{5\pi}{12}t\right) = cos(8w_0 t) + sin(3w_0 t)$$

Example 2.a: Find the fundamental frequency for the signal given

$$x(t) = \sin\left(\frac{5\pi}{6}t\right) + \cos\left(\frac{3\pi}{4}t\right) + \sin\left(\frac{\pi}{3}t\right)$$

The frequencies and periods of the three terms are, respectively,

$$w_1 = \frac{5\pi}{6}, f_1 = \frac{5}{12}, T_1 = \frac{12}{5}, w_2 = \frac{3\pi}{4}, f_2 = \frac{3}{8}, T_2 = \frac{8}{3}, w_3 = \frac{\pi}{3}, f_3 = \frac{1}{6}, T_3 = 6$$

The fundamental frequency f_0 is the GCD of f_1, f_2 and f_3:

$$f_0 = GCD\left(\frac{5}{12}, \frac{3}{8}, \frac{1}{6}\right) = GCD\left(\frac{10}{24}, \frac{9}{24}, \frac{4}{24}\right)$$

Example 3: Find the fundamental frequency of the following continuous time signal

$$x(t) = \cos\left(\frac{10}{3}t\right) + \sin\left(\frac{5\pi}{4}t\right)$$

Here the angular frequencies of the two terms are, respectively,

$$w_1 = \frac{10}{3}, \quad w_2 = \frac{5\pi}{4}$$

The fundamental frequency ω_0 should be the GCD of ω_1 and ω_2

$$w_0 = GCD\left(\frac{10}{3}, \frac{5\pi}{4}\right)$$

which does not exist since π is an irrational number and cannot be expressed as a ratio of two integers. Therefore, the two frequencies cannot be multiples of the same fundamental frequency. In other words, the signal as the sum of the two terms is not a periodic signal.